PB-VLA: Adapting Generalized VLM into a Powerful Backbone for Vision-Language-Action Model

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Abstract— Existing Vision-Language-Action (VLA) models typically rely on end-to-end fine-tuning. While effective, this paradigm compromises the inherent generalization capabilities of Vision-Language Models (VLMs) and incurs high training costs. We argue that a generalized VLM can be adapted to serve as a robust backbone for robotic manipulation. To bridge the gap between the high-level semantic understanding of VLMs and the granular requirements of robotic control, we introduce a Mixture of Layers (MoL) approach that extracts task-critical information from broad semantic features. Furthermore, to enable efficient trajectory learning under constrained model capacity, we propose a Meta Skill Module (MSM) that imposes strong inductive biases. Experiments in both simulated and real-world environments demonstrate the effectiveness of our approach. Additionally, generalization and ablation studies confirm the overall robustness of the architecture and the individual contributions of each module.

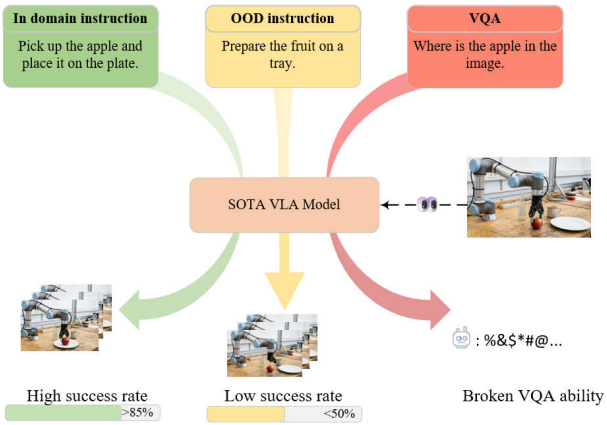

Fig. 1. The degradation phenomenon of VLM in VLA.

I. INTRODUCTION

Achieving generalizable robotic manipulation remains a cornerstone challenge in Embodied AI. Recently, the remarkable success of Vision-Language Models (VLMs) in general-purpose scene understanding [1], [2] has catalyzed the development of Vision-Language-Action (VLA) models [3], [4], which have demonstrated impressive capabilities in language-conditioned manipulation tasks.

Current VLA frameworks typically utilize pre-trained VLM backbones, fine-tuning internal parameters on large-scale robotic datasets to repurpose the model's generation capabilities from vision-language tokens to action tokens. However, this paradigm induces catastrophic forgetting [5], degrading the VLM's inherent semantic understanding and consequently limiting the VLA's generalizability. As illustrated in Fig. 1, while state-of-the-art VLA models perform well on in-domain tasks, they struggle with unseen scenarios or instructions and frequently lose the Visual Question Answering (VQA) capabilities.

We argue that preserving semantic understanding is a fundamental prerequisite for robust manipulation. Drawing inspiration from efficient adaptation strategies in VLMs [6], [7], [8], [9], we posit that a generalized VLM can be effectively adapted as the backbone for robotic manipulation without compromising its core capabilities. However, a discrepancy exists between the high-level semantic reasoning of VLMs and the fine-grained control required for robotics. To bridge this gap, we introduce the Mixture of Layers (MoL), which aggregates information across network layers to extract task-critical features. Furthermore, to enhance trajectory learning under constrained model capacity, we propose the Meta Skill Module (MSM), which leverages knowledge reuse to impose strong inductive biases. Extensive experiments in both simulated and real-world environments validate the efficacy of our approach, while visualization and ablation studies confirm the contribution of each component in overcoming performance bottlenecks.

- We demonstrate, through a novel architectural design, that a generalized VLM can be adapted into a backbone for high-performance robotic manipulation. This establishes a new paradigm for integrating VLMs into embodied agents without compromising their inherent generalizability.
- We introduce the Mixture of Layers (MoL) mechanism, which bridges the gap between high-level semantics and low-level control. This module effectively extracts fine-grained spatial information critical for manipulation from the VLM's rich semantic representations.
- We propose the Meta Skill Module (MSM) to enhance the model's representational capacity under a small action-head setting. This module leverages knowledge reuse to impose strong inductive biases, improving the model's ability to learn complex trajectories.

II. RELATED WORKS

A. Generalist Robot Policies

The pursuit of generalist robot policies has shifted the paradigm from task-specific engineering to large-scale

imitation learning. Early approaches primarily relied on Behavior Cloning (BC) [10], [11] or Reinforcement Learning (RL) [12], [13] trained on domain-specific datasets. While effective in controlled environments, these methods often struggle to generalize to unseen objects or scenes. Recent advancements have demonstrated the efficacy of Transformer-based architectures in scaling robot learning. Notable works such as Gato [14] and RT-2 [3] unify multi-modal inputs (i.e., images, text, and proprioception) into a single sequence modeling problem, enabling a single policy to perform diverse tasks. Furthermore, initiatives like Octo [15] and the Open X-Embodiment project [16] have highlighted the potential of training across heterogeneous robot embodiments. In contrast, our work adheres to the Vision-Language-Action (VLA) paradigm, employing a Large Language Model (LLM) backbone to uniformly encode multi-modal inputs for action generation. This approach allows us to directly leverage the robust generalization capabilities inherent in pre-trained VLMs.

B. Vision-Language-Action Models

VLA models are designed to integrate visual observations and proprioceptive data with natural language instructions to generate low-level robotic control signals. Capitalizing on the extensive pre-training of Vision-Language Models (VLMs), VLAs aim to transfer strong generalization capabilities to physical control. Several studies [17], [18] have reported strong performance on authoritative simulation benchmarks, such as LIBERO [19] and CALVIN [20], as well as in real-world settings. A prevailing trend involves fine-tuning the VLM backbone on large-scale robotic datasets to align representations with control tasks. While existing literature suggests that fine-tuned VLMs adapt better to manipulation tasks, this paradigm is data-intensive and often induces catastrophic forgetting, degrading the backbone's original vision-language reasoning faculties. Challenging this convention, we posit that a generalized VLM can be adapted to serve as a robust backbone for robotic manipulation.

C. Efficient Adaptation of VLMs

Given the prohibitive cost of full fine-tuning and the associated risk of catastrophic forgetting, Parameter-Efficient Fine-Tuning (PEFT) has emerged as a standard for adapting VLMs to downstream tasks. Recent works have explored Adapter-based architectures [21] to bridge the modality gap in VLA design. However, directly leveraging a generalized VLM for robotic manipulation presents distinct challenges. First, VLMs are pre-trained on internet-scale data primarily for semantic tasks such as Visual Question Answering (VQA). Extracting fine-grained, action-relevant features from the high-level representations from VLM remains difficult, as semantic richness does not imply control utility. Second, efficient adaptation methods often introduce insufficient trainable parameters, limiting the model capacity required to learn complex, diverse manipulation trajectories. To address these bottlenecks, we introduce the Mixture of Layers (MoL) to extract manipulation-critical features, and the Meta Skill Module (MSM) to enhance model capacity.

III. METHOD

In this section, we present the PB-VLA framework. We posit that a general-purpose Vision-Language Model (VLM) can be effectively adapted into a powerful backbone for robotic manipulation. We first outline the problem formulation and the overall PB-VLA architecture. Subsequently, we detail the core components: the Mixture of Layers (MoL) and the Meta Skill Module (MSM). The MoL bridges the information gap between semantic representations and detailed actions, while the MSM enhances manipulation capabilities via meta skill retrieval and action refinement.

A. Preliminary

We formulate the robot manipulation task as a conditional decision-making problem. The robot receives a visual observation $\mathcal{O}$ and natural language instruction $\mathcal{L}$, The objective is to generate a sequence of actions $\mathbf{a}$ to complete the specified task. Where the visual observation $\mathcal{O}$ typically comprises a third-person view I^{global} and a wrist-mounted view I^{wrist}.

Adopting the Vision-Language-Action (VLA) paradigm, let $\mathcal{E}_{vlm}$ denote a pre-trained VLM backbone, then the whole process can be abstracted as learning a policy π_θ:

$$\mathbf{a} = \pi_\theta(\mathcal{E}_{vlm}(I^{global}, I^{wrist}, \mathcal{L})) \tag{1}$$

B. Main Structure

As discussed in Section I, the paradigm of fine-tuning VLM on action datasets leads to poor generalization. To address this, we leverage a pre-trained VLM as a robust perceptual backbone while preserving its generalization capabilities. Furthermore, in contrast to approaches that force the VLM to directly autoregress actions, we decouple the action generation expert from the perceptual backbone. Overall, PB-VLA consists of two primary streams: Feature Extraction and Action Generation. Fig. 2 illustrates the framework of PB-VLA.

Feature Extraction. To capture rich spatial and semantic information, we process the visual observation I^{global} and I^{wrist} using DINOv2 and SigLIP simutanously, yielding visual embeddings $\mathcal{V}$. Simultaneously, the instruction $\mathcal{L}$ is tokenized into text embeddings $\mathcal{L}_{emb}$. Crucially, to explicitly extract operation-centric information from a large backbone [22], [21], we introduce a set of learnable Queries, denoted as $\mathcal{S}$. These queries are prepended to the input sequence and interact with visual and text tokens.

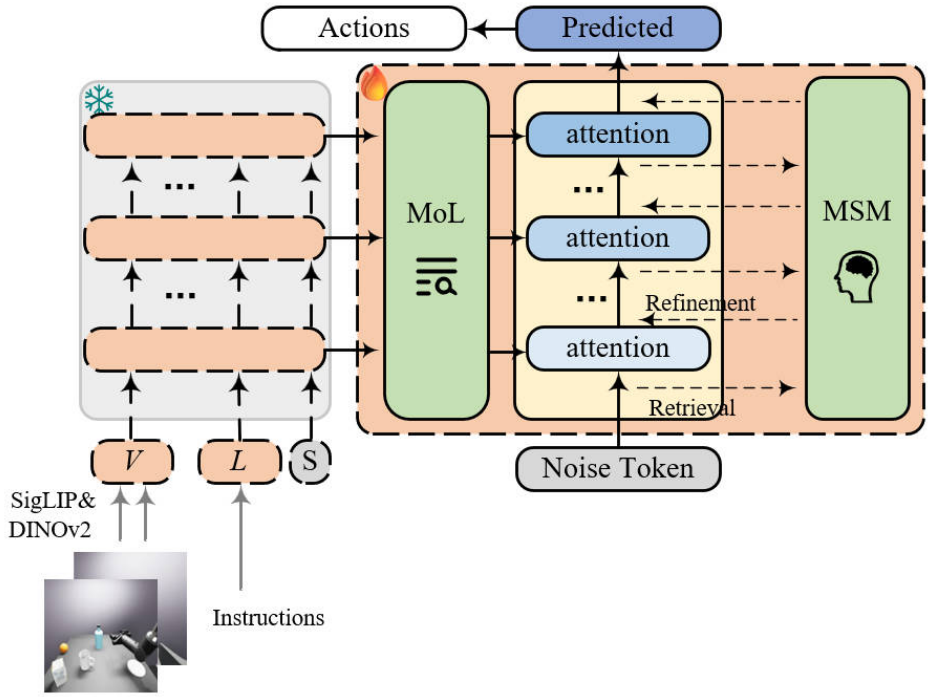

Fig. 2. Overview of the PB-VLA framework. The model extracts visual and language features, prepends learnable queries, and utilizes a pre-trained VLM as the perceptual backbone. Action generation is decoupled and performed via a denoising transformer head, with Mixture of Layers (MoL) and Meta Skill Module (MSM) enhancing feature selection and improving model capacity.

The combined input is fed into the VLM backbone. For a backbone with L layers, we extract the multi-scale hidden states:

$$\mathcal{H} = \mathcal{E}_{vlm}([\mathcal{V}, \mathcal{L}_{emb}, \mathcal{S}]) \tag{2}$$

where the output $\mathcal{H} = \{(\mathcal{V}'_l, \mathcal{L}'_l, \mathcal{S}'_l)\}_{l=1}^{L}$ represents the processed visual, text, and query features at each layer. These features serve as the condition for the subsequent action generation.

Action Generation. We design the Action Head as a multi layer denoising transformer, the action generation process is modeled as a denoising process.

When training, sample a random noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ as initial noisy action $\mathbf{a}_n$. The Action Head learns to predict the noise and reconstruct $\mathbf{a}$ conditioned on the VLM features.

$$\mathbf{a} = \mathbf{a}_n - \mathcal{D}_\phi(\mathbf{a}_n, \mathcal{H}) \tag{3}$$

where $\mathcal{D}_\phi$ is the Action Head network. The action head internally maps actions to a latent vector $\mathcal{A}$ through linear transformations for processing.

However, extracting information conducive to robotic manipulation from the VLM's massive internal representations presents significant challenges. First, the VLM extracts a vast amount of semantic and spatial information, much of which is redundant or irrelevant for low-level control. Second, the lightweight Action Head typically lacks the capacity to process these high-dimensional features for complex, diverse manipulation tasks.

To address the misalignment between VLM semantics and precise control, we introduce the Mixture of Layers (MoL) in Section III-C, which filters and selects task-relevant information from the backbone. To mitigate the capacity bottleneck of the Action Head, we propose the Meta Skill Module (MSM) in Section III-D, which retrieves and refer to meta skill from historical data to improve generalization.

Model Training. Based on the proposed architecture, we keep the parameters of VLM to prevent catastrophic forgetting, with only the parameters of Action Head and learnable query being optimized end-to-end. Following a supervised learning paradigm, the training objective is defined as:

$$\mathcal{L} = \mathbb{E}_{\mathcal{H}}[||\mathbf{a}_g - (\mathbf{a}_n - \mathcal{D}_\phi(\mathbf{a}_n, \mathcal{H}))||_1] \tag{4}$$

where $\mathbf{a}_g$ is the ground-truth action.

C. Enhancing Feature Selection via Mixture-of-Layers

The gap between VLM understanding and robotic operation introduces substantial task-irrelevant noise into the information extracted by VLM. However, within the vast amount of information extracted by VLM, there actually exists a part related to 3D manipulation [23], [24]. Therefore, if we can directly filter out this information, it would enhance subsequent action

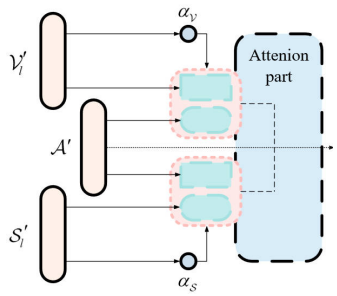

Fig. 3. Architecture of the proposed Mixture-of-Layers (MoL). MoL adaptively integrates hierarchical features extracted at various layers of the VLM via three decoupled attention branches, dynamically filtering and aggregating task-relevant knowledge for downstream action generation.

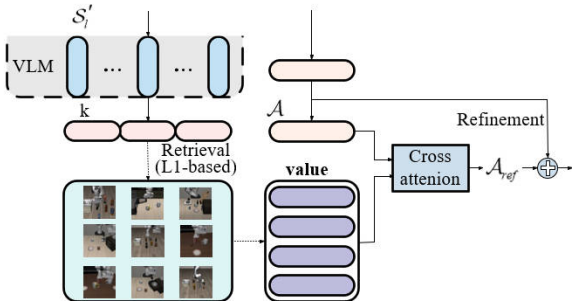

Fig. 4. Architecture of the proposed Meta Skill Module (MSM). MSM augments the Action Head with an external memory of perceptual-action key-value pairs, allowing retrieval and integration of relevant meta-skills for action refinement during inference and training.

generation. We observe that this filtering process is analogous to the dynamic integration of multi-expert information in LLMs.Inspired by this, we propose that each feature extracted by a Transformer layer in the VLM can be regarded as knowledge obtained by an expert. Drawing from the Mixture of Experts (MoE) framework, we introduce Mixture of Layers (MoL) to selectively integrate knowledge from different layers. The MoL architecture, illustrated in Fig. 3, comprises three distinct attention mechanisms or attention variants.

Specifically, for a given layer l in the Action Head, let $\mathbf{Q}_A$ denote the attention query derived from the action latent $\mathcal{A}_{l-1}$. To mitigate cross-modal interference during the softmax operation, we decouple the integration process into three parallel attention branches.

In the first branch, we leverage the processed queries $\mathcal{S}_l'$ from the VLM, which represent compressed visual-textual information. For computational efficiency, we utilize $\mathcal{S}_l'$ as the key (K_S) and value (V_S). To dynamically filter information, we compute a head-wise scaling factor $\alpha_{\mathcal{S}}$ conditioned on the $\mathcal{S}_l'$:

$$\alpha_{\mathcal{S}} = \text{sigmoid}(W_{\mathcal{S}}\mathcal{S}_l') \tag{5}$$

Subsequently, the Action Head modulates the attention mechanism via $\alpha_{\mathcal{S}}$ to selectively aggregate features.

$$\mathcal{A}_S = \text{softmax}\left(\frac{\alpha_{\mathcal{S}} Q_A K_S}{\sqrt{d_k}}\right) V_S \tag{6}$$

where d_k represents the hidden dimension of the Action Head.

Simultaneously, the second attention branch aims to incorporate layer-wise visual information $\mathcal{V}_l'$. However, given the high dimensionality of $\mathcal{V}_l'$, computing a dynamic factor would incur significant computational overhead. Therefore, we employ a learnable scalar weight α_v as a static filter to regulate the attention scores:

$$\mathcal{A}_V = \text{softmax}\left(\frac{\alpha_v Q_A K_V}{\sqrt{d_k}}\right) V_V \tag{7}$$

Finally, the third branch performs self-attention on the action latents to suppress noise:

$$\mathcal{A}_A = \text{softmax}\left(\frac{Q_A K_A}{\sqrt{d_k}}\right) V_A \tag{8}$$

where K_A and V_A are projections of the input action latent $\mathcal{A}_{l-1}$. We aggregate the outputs from these three branches to obtain the final representation:

$$\mathcal{A}_l' = \text{Mean}(\mathcal{A}_S, \mathcal{A}_V, \mathcal{A}_A) \tag{9}$$

This decoupled attention design avoids scale inconsistencies between modalities in softmax operation.

D. Enhancing Model Capacity via Meta Skill Module

Scaling laws indicate that model capacity is strictly constrained by parameter count, directly limiting generalization capabilities and overall performance [25]. Consequently, when adapting large, general-purpose VLMs to complex robotic manipulation tasks, restricting trainable parameters to solely the Action Head results in insufficient representational capacity. To address this bottleneck, we propose the Meta Skill Module (MSM), illustrated in Fig. 4. The MSM operates via two primary mechanisms: Memory Construction during training and Skill Retrieval and Action Refinement during training and inference.

Memory Construction. We maintain an external queue, denoted as $\mathcal{M}$. Each entry in $\mathcal{M}$ consists of a key-value pair. The key represents the perceptual context, formed by concatenating hierarchical features (shallow, intermediate, and deep layers) from the VLM:

$$\mathbf{k} = [\mathcal{V}_{low}' \oplus \mathcal{V}_{mid}' \oplus \mathcal{V}_{high}'] \tag{10}$$

The corresponding value is the ground-truth action sequence spanning the future chunk size. Overall, this queue constructs a meta-skill repository.

Skill Retrieval and Action Refinment. During the denoising of the acion latent, MSM leverages the meta-skill repository to refine the action latent. Specifically, MSM utilizes the multi-level features of the current observation as a query, denoted as $\mathbf{q}_{obs}$. Then, MSM retrieves the top-K (e.g., $K = 4$) nearest neighbors from $\mathcal{M}$ using the L_1 distance:

$$\mathcal{N}_K = \underset{(\mathbf{k}_i, \mathbf{v}_i) \in \mathcal{M}}{\text{Top-K}} \left(-\|\mathbf{q}_{obs} - \mathbf{k}_i\|_1\right) \tag{11}$$

The retrieved meta-skills $\{\mathbf{value}_i\}_{i=1}^K$ are projected into the latent space to serve as context $\{V_i^{\mathcal{M}}\}_{i=1}^K, \{K_i^{\mathcal{M}}\}_{i=1}^K$. Then, MSM employs a standard cross-attention mechanism to obtain the reference item $\mathbf{A}_{ref}$.

$$\mathbf{A}_{ref} = \text{softmax}(\frac{Q_A K^{\mathcal{M}}}{\sqrt{d_k}})V^{\mathcal{M}} \tag{12}$$

where the current action latent Q_A serves as the query, and the retrieved meta-skills serve as keys and values.

This retrieved reference item is integrated into the primary action flow via a residual connection, guiding the action generation with proven meta-skills.

$$\mathcal{A}_l = \mathcal{A}_l' + \mathcal{A}_{ref} \tag{13}$$

By explicitly refering to mature meta-skills, the MSM reduces the search space for the policy and improves the policy's ability to learn complex trajectories.

IV. EXPERIMENTS

To validate the hypothesis that a generalized VLM can be adapted into a backbone for high-performance robotic manipulation, we conducted experiments below. (1) We evaluate the generalization performance of PB-VLA, against existing approaches in settings requiring instruction following and unseen task generalization, showing the necessity of the proposed paradigm, (2) we evaluate PB-VLA against existing methods on standard simulation suites, demonstrating its efficiency, (3) we validate the practical applicability and robustness of PB-VLA through real-world experiments, (4) we conduct visualization and ablation studies to investigate the interpretability and effectiveness of the proposed method. All experiments are performed on 4 NVIDIA A800 GPUs.

A. Performance on Generalization Tasks

The strength of VLM and VLA lies in their ability to generalize understand language instructions and solve unseen tasks. Therefore, this paper presents its argument and constructs PB-VLA.

Instruction following. We compare the proposed method with existing approaches across tasks sharing identical visual observations but significantly differing task instructions. Specifically, each method is trained on the spatial suite of the Libero dataset for 5000 stepsand tested on its spatial suite test set. We simultaneously assess model performance using both original task instructions and synonymically rephrased instructions. The results are shown in Table I, where success rate serves as the evaluation metric.

TABLE I

Comparison of instruction following on the Libero spatial.

Method	Original Instructions	Synonymically Rephrased Instructions
OpenVLA	??	??
VLA-Adapter	98.2%	52.8%
PB-VLA	89.0%	66.2%

It can be observed that under conditions of significant language variation, most existing approaches exhibit a sharp decline in performance, whereas the proposed PB-VLA maintains the optimal success rate.

Unseen Task. In manipulation, the objects and visual conditions are varied. In this part, we test the methods' generalization when solving unseen task. Specifically, each method was trained on the Libero object suite for 5000 steps, then tested on the unseen pick-and-place task. Performance was evaluated for both original operation task and unseen task "pick wine bottle..." The results are summarized in Table II.

TABLE II

Comparison of methods on unseen task.

Method	Original Task	Unseen task
OpenVLA	??	??
VLA-Adapter	84.6%	8.0%
PB-VLA	64.8%	34.4%

We can observe that, fine-tuning methods typically encounter steep performance drops when evaluated on unseen tasks. In contrast, PB-VLA preserves its generalization capacity as it employs generalized VLM as backbone.

B. Performance in Simulation Environments

To fairly evaluate the effectiveness of PB-VLA, this section compares its task completion rates against mature baselines and state-of-the-art methods in simulation environments. We select the Libero simulation environment, training for 15000 steps and testing for 500 times. For fairness, we primarily select models utilize generalized VLM as the backbone, as the vla-adapter(frozen) and smoVLA as baselines. Concurrently, state-of-the-art models were chosen to evaluate the proposed approach's effectiveness, including [26], [17], [18], [21]. The results are presented in Table III.

The results demonstrate that PB-VLA achieves significant advantages over other methods that utilize generalized VLM as the backbone for robotic manipulation. Furthermore, PB-VLA achieves comparable performance to most state-of-the-art approaches. These findings validate the effectiveness of the proposed paradigm.

C. Performance in Real-world

To further evaluate the effectiveness of the proposed method in real-world manipulation scenarios, we conduct real-world robot experiments.

TABLE III

Comparison of methods in simulation environments. "VLM Fine-tuning" indicates whether the method requires fine-tuning the Vision-Language Model. The metrics report task success rates (%) in the "Spatial," "Object," "Goal," and "Long" suites, as well as average performance and parameter counts.

Method	VLM Fine-tuning	Spatial	Object	Goal	Long	Avg.	Params (B)
OpenVLA	Yes	84.7	88.4	79.2	53.7	76.5	7.0
molmoAct	Yes	87.0	95.4	87.6	77.2	86.6	7.0
ThinkAct	Yes	88.3	91.4	87.1	70.9	84.4	7.0
SmolVLA	No	93.0	94.0	91.0	43.6(?)	??	2.2
VLA-Adapter (Frozen)	No	94.4	94.2	86.8	11.6 (?)	??	0.5
PB-VLA	No	95.6	97.0	93.6	49.4 (?)	??	0.3

Experimental Setup. All experiments are deployed on an AgileX PiPer 6-DoF robotic manipulator equipped with a parallel gripper, resulting in a 7-DoF system (illustrated in Fig. 5). Visual observations are acquired via 2 RGB cameras: a static front-view camera (main camera) and a wrist-mounted camera (wrist camera).

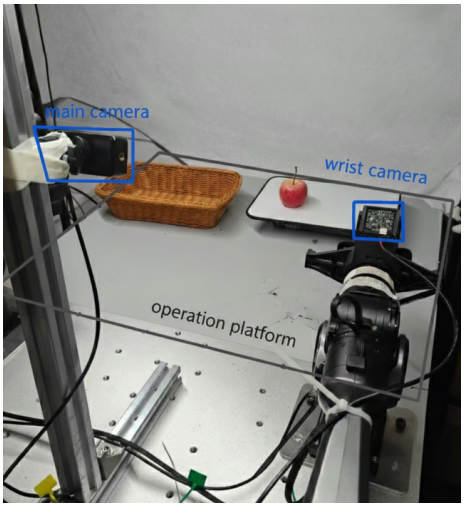

Fig. 5. Experimental setup for real-world manipulation tasks.

Evaluation Tasks. To assess real-world performance and generalization capabilities, inspired by recent works [27], [28], we design four categories of evaluation tasks: (1) pick the apple and place it in the basket, (2) pour water into the bowl, (3) pick with instruction following, and (4) pick with unseen task.

Specifically, the pick with instruction following task evaluates language generalization by providing synonymically rephrased instructions not seen during training. The pick with unseen task setting evaluates task generalization ability. Under this setting, the model is trained exclusively on pick-and-place tasks with apples but is evaluated on novel target objects (a water bottle or a toy duck) accompanied by corresponding novel instructions.

Data Collection and Training. For each task, we collect 50 demonstration trajectories via teleoperation. SmolVLA[29], VLA-Adapter (Frozen) [21], VLA-Adapter [21] and PB-VLA are trained for 5000 steps.

Results and Analysis. Each task is evaluated with 20 real-world rollouts. The success rates are reported in Table IV.

TABLE IV

Real-world Task Successes (Number of successful rollouts out of 20 trials for each task).

Method	Pick apple into basket	Pour water into bowl	Instruction following	Unseen task
smolVLA	5	6	5	4
VLA-Adapter (Frozen)	2	0	1	1
VLA-Adapter	12	10	11	0
PB-VLA	17	18	16	15

Based on the results above, we can draw the following conclusions:

- On standard manipulation tasks, our proposed PB-VLA achieves a high success rate, outperforming all baseline methods.
- Under the pick with instruction following setting, PB-VLA demonstrates superior robustness to language variations compared to existing approaches.
- On unseen tasks, while conventional fine-tuning paradigms suffer from catastrophic performance degradation, methods leveraging generalized VLMs as backbones (i.e., VLA-Adapter (Frozen) and PB-VLA) exhibit stronger generalization without steep performance drops. Furthermore, due to the architectural designs that are specifically customized to adapt generalized VLM backbones, PB-VLA achieves significantly higher success rates.

Qualitative observations from the pick with unseen task setting provide further insights. Existing paradigms, such as smolVLA and VLA-Adapter, are prone to overfitting on the training trajectories. Specifically, we observe that the VLA-Adapter frequently exhibits erratic end-gripper oscillations when approaching novel targets, ultimately leading to task failure. We hypothesize that this instability stems from a severe covariate shift in the wrist-camera observations compared to the training data. In contrast, the generalized VLM backbone employed in our approach consistently extracts invariant manipulation features across novel objects, allowing PB-VLA to

maintain robust execution and high success rates.

D. Visualization Analysis

To investigate whether the proposed strategy effectively overcomes bottlenecks encountered during adapting VLM backbone, this section conducts 2 visualozation analysis. First, we verify whether MoL successfully identifies layer experts proficient in spatial information from the massive VLM data. Then, we discusses whether key selection in MSM can effectively retrieve approximate skill.

Based on a trained model, when processing the "pick black bowl..." task, the corresponding $\alpha_{\mathcal{S}}$ values for each layer are plotted in Fig. 6.

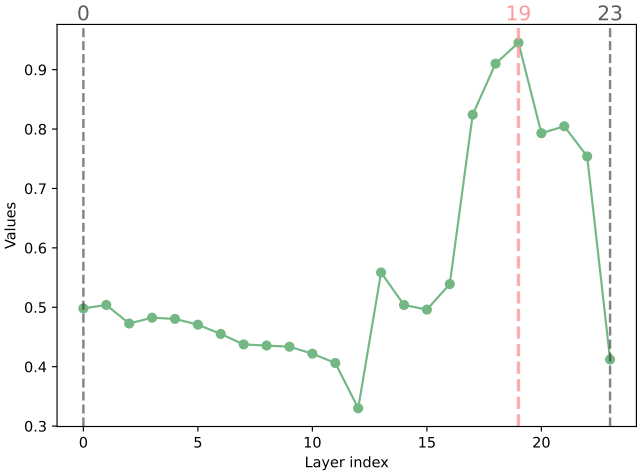

Fig. 6. Visualization of the $\alpha_{\mathcal{S}}$ values for each layer.

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In shallow layer 0, intermediate layer 19, and deep layer 23, the attention heatmap of instruction over observed images are visualized as Fig. 7.

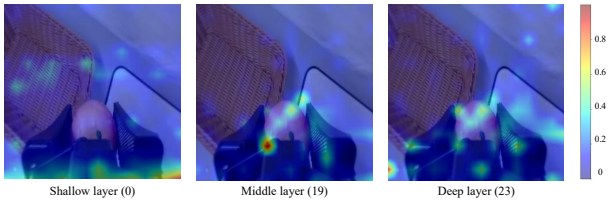

Fig. 7. Visualization of the attention map for language instruction at shallow layer 0, middle layer 19 and deep layer 23.

Although the VLM cannot directly provide the visual information required for robotic operations. Howecver, we can observe from the experiments above, the MoL successfully identifies operation-relevant visual information from vast information.

Next, we visualize the L1 distances between a specific Key and other Keys within the MSM during model inference, alongside the L1 distances between Values. The cosine distances are also visualized, as shown in Fig. 8.

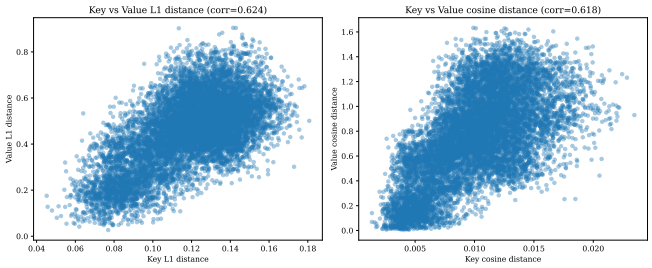

Fig. 8. Visualization of the correlation using L1 distances and cosine distances between Keys and between Values within the MSM.

It can be observed that the distances between Keys and between Values exhibit a significant positive correlation, the correlation coefficients = 0.624 and 0.618 respectively. This indicates that the Key construction scheme based on 3-layer VLM information within the MSM successfully constructs Keys that query useful skills. Simultaneously, L1 distance is more effective than cosine distance in distinguishing differences among meta-skills.

E. Ablation Studies

To further valide the significance of each proposed strategy in overcoming bottlenecks during adapting VLM. We conduct ablation studies of the proposed MoL and MSM on Libero spatial suite in Table V. All models are trained for 5000 steps.

TABLE V

Ablation study of MoL and MSM.

Method	MoL	MSM	success rate (%)
PB-VLA (w/o MoL, w/o MSM)	–	–	46.6
PB-VLA (w/ MoL only)	✓	–	67.4
PB-VLA (w/ MSM only)	–	✓	66.2
PB-VLA (full)	✓	✓	75.0

It can be observed that PB-VLA performs poorly without MoL and MSM. The MoL assists the Action head in filtering beneficial information for robotic manipulation from the massive information from VLM, achieving significant performance improvements. Simultaneously, the MSM enhances the model's capacity with limited parameters, enabling it to handle diverse scenarios.

V. Conclusion

In this work, we introduce PB-VLA, a novel architecture design with novel paradigm that adapts generalized Vision-Language Models (VLMs) as powerful backbones for Vision-Language-Action (VLA) models. By incorporating the Mixture of Layers (MoL) mechanism to extract task-critical cues and the Meta Skill Module

(MSM) to enable efficient trajectory learning with limited parameters, PB-VLA achieves remarkable generalizing performance on instruction following and unseen tasks. In the future, we will configure multiple heads customized to different tasks and bodies. By employing a single brain with multiple specialized cerebellums, we aim to construct a universal VLA model.

References

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